Lorenzo Masoero

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SUMMARY

Senior Applied Scientist at Amazon Weblab (Amazon's largest A/B testing platform). Research focus on causal inference, experimental design, and agentic AI for experimentation—with publications in JRSS-B, Statistical Science, NeurIPS, Biometrika, and JASA. PhD from MIT (EECS), supervised by Prof. Tamara Broderick.

EXPERIENCE

Senior Applied Scientist III Amazon, Weblab 2021–Present

Online Experimentation with Interference. Pioneered "multiple randomization designs" (joint work with Prof. Guido Imbens and Prof. Thomas Richardson, amongst others)—experiments that jointly randomize across multiple marketplace units to detect spillover effects. Published in JRSS-B (2026) and Statistical Science (2023).

Advancing Experimentation Methodology. Deployed a distributed PySpark engine for covariate-adjusted causal estimation (KDD 2023) serving thousands of daily experiments. Developed Demonstration Experiments (with Vijaykumar & Imbens) and robust switchback designs (with Missault) for time-randomized experiments.

Agentic Workflows for Experimentation. Built a multi-agent system for automated documentation maintenance (TriAgent, IAAI 2027) and autonomous LLM experimenters for A/B test analysis (AMLC 2026 Workshop).

EDUCATION

PhD, Electrical Engineering and Computer Science MIT, 2016–2021
Advisor: Prof. Tamara Broderick. Thesis on Bayesian nonparametric methods for genomic variant discovery.

MA, Statistics and Applied Mathematics (with distinction) Collegio Carlo Alberto, 2015–2016

MA, Quantitative Finance (110/110 magna cum laude) Università di Torino, 2015–2016

BA, Economics (110/110 cum laude) Università di Torino, 2012–2014

SELECTED PUBLICATIONS

- Multiple Randomization Designs: Estimation and Inference with Interference. JRSS-B, 2026.
- Multi-Armed Bandits with Network Interference. NeurIPS 2024.
- Experimental Design in Marketplaces. Statistical Science, 2023.
- More for Less: Predicting and maximizing genetic variant discovery via BNP. Biometrika, 2022.
- Scaled process priors for new-feature counts via random scaling in BNP. JASA, 2022.
- Online Activity Prediction via Generalized Indian Buffet Process Models. Annals of Applied Statistics, to appear.
- Mind the Gap: A Deployed Multi-Agent System for Documentation Maintenance. IAAI 2027.
- Can AI Agents Simulate A/B Test Outcomes? WAB@COLM 2026; CODE@MIT 2026.
- Demonstration Experiments. Pre-print, 2025.
- From Offline Evidence to Online Action. ICML 2026 Workshop.

AWARDS

SBSS Best Student Paper Award, American Statistical Association (2020) · Andrew & Erma Viterbi Fellowship (2017)

SERVICE

Reviewer for JRSS-B, JASA, JMLR, NeurIPS, KDD, AISTATS, ProbML (2018–2026).

SKILLS

Python (NumPy, SciPy, Pandas, PySpark, scikit-learn) · Bayesian inference & causal inference · LaTeX